



HISTORY

UNIT

1

Evolution of Humans and Society - Prehistoric Period

Learning Objectives

- To trace the world's early history
- To understand human evolution
- To know prehistoric Tamilagam up to the Iron Age



Introduction

We live in the age of Information Technology. The mobile phones have literally put the world on our finger tips. The all-encompassing knowledge that we possess now, which has helped in the development of powerful technology, did not emerge all of a sudden. The foundation of our modern life was facilitated by the development of the process of cognition among the human ancestors in the prehistoric age.

Prehistoric people were the pioneers of creative knowledge. From the artefacts and the languages they developed, we are able to understand how intelligent they were.

Palaeoanthropologists and archaeologists excavate the soil and rock layers on the earth and extract evidence about human ancestors. These layers and the fossils are scientifically dated to study the various stages in human evolution and prehistory. Through the gathered evidence, they attempt to understand the evolution of human history and developments in a chronological order.

Archaeology is the study of human past through the analysis and interpretation of material remains.

Palaeoanthropology is the study of the human ancestors and their evolution by the study of the fossil remains.

The earth was formed approximately 4.54 billion years ago. Gradually, conditions emerged for the growth of organisms. Then plants and animals came into being, and thereby foundation was laid for the evolution of humans. The long span of time in the history of earth is divided into eras, periods and epochs by the geologists

1 billion = 100 crore

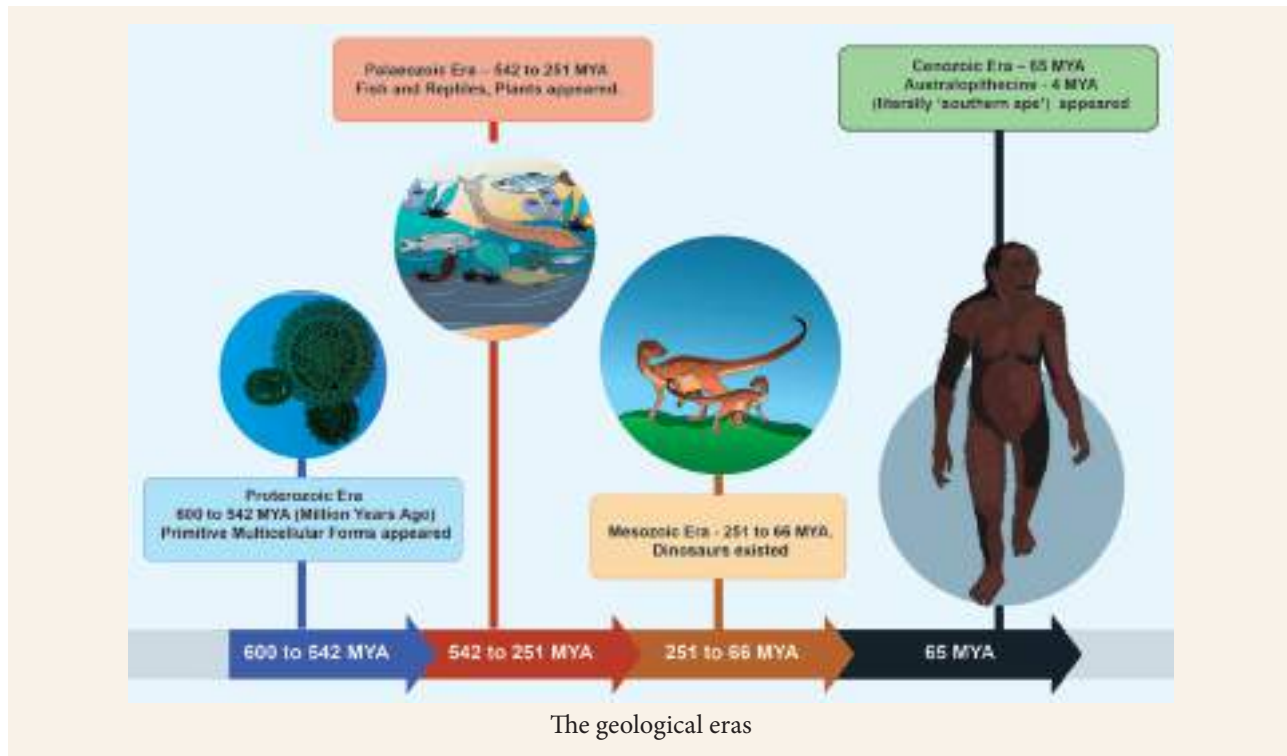
1 million = 10 lakh

Australopithecines were the apes from which modern humans evolved. Now they are extinct, but they are considered to be the close relatives of humans.

1.1 Origin of the Earth and the Geological Ages

The history of humans is closely related to the history of the earth. The earth contains geological, archaeological and biological records of historical times in its upper layers. They are important for reconstructing the history of the earth and various living organisms. The fossil bones of the human ancestors are embedded in the earth's layers.





1.2

Human Enquiries into the Past and Origin of the World

The Age of Speculation

Humans are the only species on earth concerned with understanding as well as explaining the world and the universe. In the course of evolution, humans became conscious and knowledgeable. They turned curious and began to think and ask questions about nature, organisms and the world around them. At first, they considered nature as God. They worshipped sun, moon and various natural forces about which they developed their own understanding, some of which was not scientific. The lack of scientific knowledge about the creation of the world is reflected in the ancient writings and religious literature.



BC = (BCE) – Before Common Era

AD = (CE) – Common Era

MYA – Million years ago

Scientific Foundations of Geology, Biology and Archaeology

The beginning of history writing can be traced to the ancient Greeks. Herodotus (484–425 BC (BCE)) is considered the Father of History, because the history he wrote was humanistic and rationalistic. The rise of scientific inquiry into the origin of humans was possible because of

- The interest in collection of archaeological remains and the opening of museums after the Renaissance Movement.
- The development of ideas of stratigraphy and geology.
- Charles Darwin's theory of biological evolution.
- The discovery of human and animal fossils, stone tools, and artefacts of early civilizations.
- The ability to decipher early scripts.

Stratigraphy – The study of origin, nature and relationships of rock and soil layers that



were formed due to natural and cultural activities.

Oldest Museum of the World – The museum of Ennigaldi- Nanna in Mesopotamia was established in 530 BC (BCE). The Princess Ennigaldi was the daughter of the Neo-Babylonian king Nabonidus. The Capitoline Museum in Italy is perhaps the oldest surviving museum (1471 AD (CE)) at present. Ashmolean Museum at Oxford University is the oldest university museum in the world. It was established in 1677 AD (CE).

Herbert Spencer's (1820–1903 AD (CE)) biological evolution, and Charles Darwin's (1809–1882 AD (CE)) theory on concepts of **natural selection and survival of the fittest** contributed to the scientific understanding of human origins. Charles Darwin published the books *On the Origin of Species* in 1859 and *The Descent of Man* in 1871.

Natural selection – The process by which organisms that are better adapted to their environment would survive and produce more offspring.

Survival of the fittest means “survival of the form that will leave the most copies of itself in successive generations.”

Fossil – Prehistoric animal or plant that turns into stone over a period of time (millions of years) because of chemical and physical processes. Animal bones are preserved due to mineralization. Palaeontology is the study of fossils.

Stone Age – the period when stone was mainly used for making implements.

Bronze Age – the period when bronze metallurgy (extraction of metal from ores) developed.

Iron Age – the period when iron was smelted to produce implements.

Since the 19th century, scholars have used advanced scientific techniques. They undertook systematic studies to contribute to

the current state of knowledge on prehistory, human origins and the early civilisations. Now the theory of human evolution is widely accepted.

1.3 Prehistory: From Australopithecus through Homo erectus to Homo sapiens

Who are we? What is the name of our species?

We are *Homo sapiens*, the species of modern humans.

Human Evolution and Migration

The chimpanzee, gorillas and orangutans, along with humans, are collectively called the Great Apes. Among them, the chimpanzee is genetically the closest to humans.

The ancestors to humans were called Hominins, and their origins have been traced in Africa. They evolved from those origins and then began to move to other parts of the world in due course of time. The Hominins emerged around 7 to 5 million years ago. Skeletons of Australopithecus, one of the early species of this tribe, have been found in Africa.

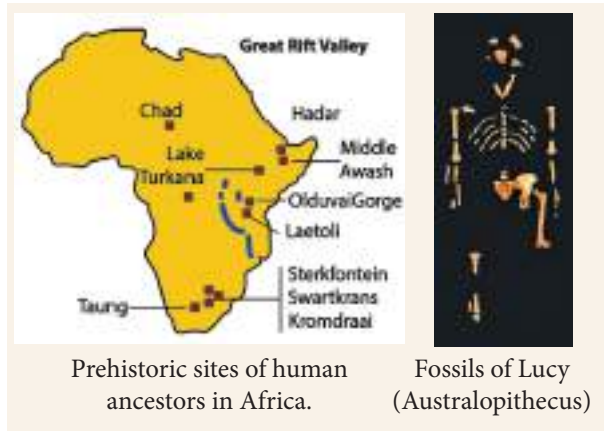
The Great Rift Valley in Africa has many sites that have evidence for the prehistoric period.



The DNA of a chimpanzee is 98% identical to that of a human being.

The Great Rift Valley is a valley-like formation that runs for about 6,400 km from the northern part of Syria to Central Mozambique in East Africa. This geographical feature is visible even from the space, and many prehistoric sites are found in eastern Africa.

Human ancestors are divided into various species according to their physical features.



Hominid refers to all the species of the modern and extinct great apes, which also includes humans..

Hominins (a zoological tribe) refers to the close relatives of human ancestors and their sister species including *Homo sapiens* (the modern humans) and the extinct members of *Homo neanderthalensis*, *Homo erectus*, *Homo habilis* and various species of *Australopithecines*. Humans are the only living species of this 'tribe'. They stand erect, walk with two legs and have large brains. They can use tools and a few of them can communicate. It excludes the gorillas.

Homo habilis (handy human) was the earliest known human ancestors to make tools in Africa about 2.6 million years ago. Around 2 million years ago, the species of *Homo erectus/ergaster* emerged. This species made hand axes between 2 and 1 million years ago. They began to spread into various parts of Asia and Africa in time.

Anatomically, modern humans, called *Homo sapiens* (wise man), first appeared around 3,00,000 years ago in Africa. It is believed that these modern humans eventually migrated and dispersed into various parts of the world from around 60,000 years ago.

Prehistoric Cultures

Prehistoric period does not have evidence of writing. While the fossil bones are classified as various species such as *Homo habilis*, *Homo erectus* and *Neanderthalensis*,

based on the lithic tools, cultures are assigned names such as Earliest Lithic Assemblages, Oldowan Technology; Lower, Middle and Upper Paleolithic; and Mesolithic cultures.

Period



The chimpanzee and the pygmy chimpanzee (also known as Bonobo) are our closest living relatives.

Earliest Lithic Assemblages of Human Ancestors

The earliest tools made by human ancestors are found in Lomekwi in Kenya. They are dated to 3.3 million years ago. Oldowan tools occur in the Olduvai gorge in Africa. They are 2 to 2.6 million years old. The human ancestors (*Australopithecines*) used hammer stones and produced sharp-edged flakes. The tools were used for cutting, slicing and processing food.

Lower Paleolithic Culture

The Lower Paleolithic Culture is marked by the human ancestors belonging to the species *Homo habilis* and *Homo erectus*. The human ancestors flaked large stone blocks and designed various tools including hand axes. These tools, which are found in Africa, Asia, and Europe, are dated the earliest to about 1.8 million years ago. They made various tools such as hand axes and cleavers to meet their subsistence needs. These tools are also known as bifaces. These tools have physical symmetry and convey the humans' cognitive (perception) skills. This culture is called the Lower Paleolithic Culture. The hand axe tools are also known as Acheulian. This tool-making tradition continued from 250,000 years to 60,000 years ago in India.

Acheulian – They were first hand axes recognized at a place called St. Acheul in France. Hence they are called Acheulian tools.

Bifaces are tools that have flaking on both sides (bi = two, face = side).





Stone tools from Kenya
about 2.3 million years old



An Olduvai chopper

Subsistence necessities of prehistoric humans were mainly food and water.

The human ancestors perhaps did not possess complex language skills as we have now. They might have voiced a few sounds or words and possibly used sign language. They were intelligent enough to select stones as **raw material** and used the **hammer stones** to carefully flake the rocks and design tools for their needs. They hunted animals, fed on the meat of the animals killed by predators and gathered plant foods such as roots, nuts and fruits. In India, the Acheulian tools have been found near Chennai and many other sites such as Isampur in Karnataka and Bhimbetka in Madhya Pradesh.

Raw material is the naturally available stone block or pebbles selected by humans for making tools. Since these stones produced flakes with sharp edges, they were selected for making stone tools.

Core is the main block of stone from which small chips are flaked by using a hammer stone.

Flake is a small chip removed from a large stone block called the core.

Middle Paleolithic Culture

After about 3,98,000 years Ago, further changes took place in the **lithic technology** in Africa. The *Homo erectus* species existed during this period. Anatomically modern humans are said to have emerged around 3,00,000 years ago.

Lithic Technology: 'Lith' means stone. The methods and techniques involved in the

production of stone tools are called Lithic technology.

The hand axes turned out to be much attractive in design and many smaller tools were also produced. The core was prepared and then tools were made. Points and scrapers were used. Short blades were also produced. The lithic tool-making tradition of the **Levalloisian** belonged to this period. The tools made during this time are found in Europe and Central and Western Asia.

Levalloisian tools are the implements made after preparing the core. It was named after the town of Levallois in France.

The Middle Paleolithic Culture appeared between 3,85,000 and 1,98,000 years ago in Europe and parts of Western and South Asia. The tools that were made during this period were in use till about 28,000.

The people of this period were called Neanderthals. They buried the dead people systematically.

Upper Paleolithic Culture

The cultural phase that succeeded the Middle Paleolithic is called the Upper Paleolithic phase. This period was marked by innovation in tool technology. Long blades and burins were produced during this time. People used different varieties of silica- rich raw materials in this phase. Numerous paintings and art objects were made. The diversity of artefacts suggests the improvement in cognitive skills and the development of languages. Microliths appeared in this phase.

Burin is a stone-made chisel with a sharp cutting edge.

The modern humans, who first appeared as a result of human evolution in the sub-Saharan Africa 300,000 years ago, began to move to various parts of Asia around 60,000



Hand axe -
London Museum



Flint biface from
Saint-Acheul, France



A cleaver



Omo Kibish point



Middle Paleolithic
flakes and tools, India

years ago. They probably replaced the earlier populations. In Europe, humans known as Cro-Magnons lived in this period.

Horns and ivory were used for making tools and art works. Bone needles, fishhooks, harpoons and spears were also employed creatively. The humans of this time wore clothes and cooked food. The dead were placed in the burials with folded hands placed over their chest. Pendants and richly carved tools were also seen in use. Evidence from paintings, clay model sculptures and carvings is available. Images of Goddess Venus were made of stones and bones in Europe and in some parts of Asia.

Ice Age – the period before 8,000 BC (BCE) when many parts of the world remained covered by ice sheets and snow.



Lascaux – Rock painting from
West France – 17,000 years old

Mesolithic Culture

Mesolithic period is known as the Middle Stone Age, as it is placed between the Paleolithic and Neolithic periods. People mainly used microlithic (small stone) tools

during this period. These people were hunter-gatherers. With the global warming occurring after the Ice Age, they became highly mobile and occupied various eco-zones (coastal, hilly, riverine and dry region).

People of Mesolithic period widely employed microlithic technology. They made tiny artefacts that were less than 5 cm in size. They produced points, scrapers and arrowheads. They also used geometric tools such as lunates, triangles and trapezes. These tools were hafted onto wooden or bone handles and used.

Microliths are stone artefacts of small size.



Rock paintings from Bhimbetka

Neolithic Culture and the Beginning of Agriculture

The period called Neolithic marks the beginning of agriculture and animal domestication. It is an important phase in history. Early evidence

of the Neolithic period is found in the fertile crescent region of Egypt and Mesopotamia, the Indus region, the Gangetic valley and in



China. By about 10,000 BC (BCE) to 5000 BC (BCE), agriculture had come to be practised in these regions.



Wheat, barley and peas were cultivated around 10,000 years ago. Fruit and nut trees were cultivated around 4,000 BC (BCE). They comprised olives, figs, dates, pomegranates and grapes.

Fertile Crescent Region refers to the area covering Egypt, Israel-Palestine and Iraq, which is in the shape of crescent moon.

Neolithic Age is called the 'new age', because of the new grinding and polishing techniques used for the tools. The Neolithic people also used the flaked stone tools. Until the Mesolithic period, people mainly hunted and gathered food for their subsistence. By hunting and gathering people obtained very limited food as a result of which only a small number of people could exist in a particular region.

The introduction of domestication of animals and cultivation of plants at home led to production and supply of large quantities of grains and animal food. The fertile soil deposited by the river on its banks helped the growth of agriculture. People preferred to live on river banks as it was better for adaptation. As a result of domestication and cultivating plants, there was an surplus food production. The surplus food production was a main factor for the development of early civilisations. Permanent residences were built and large villages emerged as a result. Hence, the development of this period is called Neolithic Revolution.

1.4 Prehistoric Tamilagam

Lower Paleolithic Culture in Tamil Nadu

One of the oldest Stone Age tools in the world made by human ancestors, called

hominins, had been produced in Tamil Nadu. These stone tools are found near the Chennai region at several sites, especially at Athirampakkam. The **archaeological excavations** at this site and **cosmic-ray exposure dating** of the artefacts suggest that people lived here about 1.5 to 2 million years ago. The Kosasthalaiyar River is one of the major cradles of human ancestors in the world. The people who lived here belonged to the species of *Homo erectus*.

Archaeological excavation refers to digging undertaken to recover archaeological evidence such as stone tools, pottery, animal bones and pollens, in order to understand the past lifestyle of humans.

Cosmic-ray exposure dating – method in which exposure to cosmogenic rays is done for dating the samples.

In 1863, Sir Robert Bruce Foote, a geologist from England, first discovered Paleolithic tools at Pallavaram near Chennai. They are the earliest finds of such tools in India. Hence, the hand axe assemblages were considered as the **Madras Stone Tool Industry**. The tools that he discovered are now housed in the Chennai Museum.

The Paleolithic people hunted wild animals and gathered the naturally available fruits, roots, nuts and leaves. They did not have knowledge of iron or pottery making, which developed much later in history.

Hand axes and cleavers are the important tool types of the Lower Paleolithic period. These tools fitted with a wooden and bone handle and were used for cutting, piercing and digging. The people of this time used hammer stones and spheroids. The quartzite pebbles and cobbles were chosen as raw materials. The tools are found in the soil deposits and also in the exposed river side. They were found at Pallavaram, Gudiyam cave, Athirampakkam, Vadamadurai, Erumaivettipalayam and Parikulam.



The Lower Paleolithic tools are also found in the North Arcot and Dharmapuri districts. The people belonging to this period used **basalt rocks** for manufacturing artefacts. However, the southern part of Tamil Nadu and Sri Lanka do not have evidence of Lower Paleolithic Culture.

Basalt rocks are igneous rocks: Igneous rocks are those formed from the molten lava from the earth.

The Lower Paleolithic Culture is datable to about 2 - 1.5 million years at Athirampakkam. This cultural phase continued in other parts of India up to 60,000 years ago.

Middle Paleolithic Culture in Tamil Nadu

In the course of time, the **Middle Paleolithic Culture** emerged during 3,85,000 - 1,72,000 years ago. The tool types of this period underwent a change and smaller artefacts were used. Cores, flakes, scrapers, knives, borers, Levalloisian flakes, hand axes and cleavers are the artefact types of this period. Compared to the previous phase, these tool types became smaller in size.

Evidence for the Middle Paleolithic Culture can be observed in some parts of Tamil Nadu. In the southern part of Tamil Nadu, at T. Pudupatti and Sivarakkottai, artefacts of the Middle Paleolithic tools have been collected. Also near Thanjavur and Ariyalur, similar artefacts have been found.

Mesolithic Culture in Tamil Nadu

In many parts of the world, and in some parts of India, the Upper Paleolithic Culture succeeded the Middle Paleolithic Culture. There is no evidence for the Upper Paleolithic Culture in Tamil Nadu. But the people who used microliths or small-stone artefacts lived in many parts of Tamil Nadu. Since this cultural period occurred between Paleolithic and Neolithic Culture, it is known as Mesolithic Culture or Middle Stone Age.



Microlithic flakes from
Tamil Nadu

Evidence for the existence of Mesolithic hunter-gatherers is found at Chennai, North Arcot, Dharmapuri, Salem, Coimbatore, Ariyalur, Tiruchirappalli, Pudukkottai, Madurai, Sivaganga, Tirunelveli and Kanniyakumari. The teri sites near Thoothukudi have evidence of microlithic artefacts. These sites have red sand dunes called **teris**.

The people of this period used small artefacts made of **chert** and **quartz**. The tool types are **scrapers**, **lunates** and **triangles**. These people hunted wild animals and gathered fruits, nuts and roots for their **subsistence**.

Scrapers are tools used for scraping the surfaces. Scrapers are similar to the tools used in the kitchen for removing skin of vegetables.

Triangles are tools in the shape of triangles.

Lunates are tools in the shape of a crescent.

Neolithic Culture in Tamil Nadu

The culture that domesticated animals and cultivated crops is called Neolithic. It is known as the New Stone Age. The Neolithic people used polished stone axes called celts. Cattle rearing was their main occupation. They lived in small villages with houses made of thatched roof and walls plastered with



Timeline: The Course of Cultures in Ancient Tamilagam		
Culture	Time Period	Cultural Traits
Paleolithic Period	Circa. 20,00,000 years to circa. 8,000 BC (BCE)	Hand axes, cleavers Hunting and gathering
Mesolithic Period	Circa. 8,000 years to circa. 1,300 BC (BCE)	Microlithic tools No knowledge of metal Hunting of animals and birds Gathering of plant food
Neolithic Period	Circa. 2,000 BC (BCE) to 1,000 BC (BCE)	Polished Stone Axes Microliths Domestication of animals Cultivation of crops Multiplicity of groups Co-existence of hunter-gatherers and pastoral groups
Iron Age	Circa. 1,300 BC (BCE) to 500 BC (BCE)	Megalithic burial custom Co-existence of hunter-gatherers and pastoral groups Development of chieftdom Knowledge of iron, black and red ware, black ware ceramics Craft specialisation, specialised groups: potters, blacksmiths
Early Historic and Sangam Age	300 BC (BCE) to 300 AD (CE)	Cultural traits of Iron age Monarchies of Chera, Chola and Pandya Development of hero worship Poetic traditions and literature Trade and exchange by sea

clay. Evidence of Neolithic village is found at Payyampalli in Vellore district and a few sites in the Dharmapuri region.

DO YOU KNOW? Neolithic people perhaps devised the first pottery. They made pottery, using a slow wheel called turn-table or made pottery out of hand. Before firing, the pottery was polished with pebbles. This process is known as burnishing.

Payyampalli is a village in Vellore district of Tamil Nadu. The earliest evidence for the domestication of animals and cultivation of plants is found at this site, which was excavated by the Archaeological Survey

of India. Evidence for pottery making and cultivation of horse gram and green gram has been found in this village.

Iron Age/Megalithic period

The **cultural period** that succeeded the Neolithic is called the Iron Age. As the name suggests, people used iron technology. It preceded the Sangam Age. The Iron Age was a formative period and the foundation for the Sangam Age was laid in this time. During the Iron Age, many parts of Tamil Nadu were occupied by people. An exchange relationship developed among the people.

The people of this age had knowledge of metallurgy and pottery making. They used iron and bronze objects and gold ornaments.



Lemuria and the Tamils

Some researchers relate the origin of the Tamils to the submerged continent of Lemuria. This theory of Lemuria continent was proposed in the 19th century. In the wake of advancements in plate tectonics theory, differing views are put forth by scholars.

The available literary references point to the submergence of areas around Kanniyakumari. Some parts of Sri Lanka and Tamil Nadu were connected by land about 5000 years BC (BCE). It is possible that some land might have submerged near Kanniyakumari and around the coast of India, because of the rising sea levels. Underwater surveys are necessary in this area.

Archaeological research reveals that at least a section of people may have been living continuously in South India, including Tamil Nadu, from the Mesolithic and Neolithic times.

They used shell ornaments and beads made of carnelian and quartz. The evidence for Iron Age is found at many sites including Adhichanallur in Thoothukudi district, Sanur near Madhuranthakam and Sithannavasal near Pudukkottai. Megalithic burial sites are found in the whole of Tamil Nadu.

Megalithic Burial Types

The Iron Age is also known as megalithic, since people created burials with large stones for the dead people. Within these burials, the skeletons or a few bones of the dead persons were placed along with grave goods including iron objects, carnelian beads and bronze objects. Some of the burials do not have human bones and they have only the grave goods. They may be called 'Memorial Burials'.

Grave goods are the objects placed in the burials along with the physical remains (bones) of the dead. People may have believed that these would be useful in the after-life. Egyptian pyramids also have similar artefacts.

Similar burials were also built in the early historic period or the Sangam Age. The Sangam literature mentions the various burial practices of the people. The megalithic burials are classified as dolmens, cists, menhirs, rock-cut caves, urn burials and sarcophagus. The burial types of Kodakkal (umbrella stone), Toppikkal (hatstone) and Paththikal (hoodstone) are found in Kerala.



Sir Robert Bruce Foote discovered the first Paleolithic tools in India at Pallavaram



Tools discovered by Robert Bruce Foote



Gudiyaam Cave near Chennai

Dolmens, table-like stone structures, were erected as funerary monuments. Cists are stone enclosures buried under the earth. They were created by placing four stone slabs on the sides, one on top of each other. The cists and dolmens have openings called portholes. Urns are pottery jars and were used for burying the dead. Sarcophagi are burial receptacles made of terracotta. They sometimes had multiple legs. Menhirs are pillar-like stones erected as part of the burials or memorials.

Portholes are holes found in the cists and dolmens on one side. They may have acted as the entrance to the burials. There is a view that they were meant for the movement of the soul or spirit.



A Menhir



An urn covered with lids
in Adhichanallur



A stone circle with Menhir
and the capstone



Sarcophagus



Dolmen



Cist, Kodumanal



Kodakkal, Kerala



Toppikkal, Kerala



Paththikal, Kerala

The menhirs may have been erected for the heroes in the Iron Age. The tradition of hero stones might have begun in the Iron Age or even before.

Agriculture and Pastoralism

The people in the Iron Age practised agriculture, domesticated cattle and sheep, and some of the groups were hunting and gathering. Millets and rice were cultivated. Irrigation management developed in this period, since many of the megalithic sites are found nearby rivers and tanks. In the deltaic regions, irrigation as a technology had developed. Evidence of rice is seen in the megalithic sites like Adhichanallur in Thoothukudi district and Porunthal near Palani.

Iron Age Society and Polity

The Iron Age society had farming communities, pastoralists and hunter-

gatherers. Craft specialists, potters and blacksmiths were the professionals during this period. The society had several groups of people (tribes). The size of the burials and the variations found in the burial goods suggest the existence of numerous social groups and their diverse practices. Some of them seem to have had organised chiefdoms. Cattle lifting leading to wars and encroachment and expansion of territories had also started taking place in this period.

Pottery

Pottery is an important evidence found in the archaeological sites. The Iron Age and Sangam age people used the black and red colours to make black ware and red ware pottery. Potteries were used for cooking, storage and dining purposes. The black and red ware pottery has a black inside and a red outside, with lustrous surfaces.



Black and Red ware in Adhichanallur.



(a) Iron sword and dagger in Adhichanallur
(b) Bronze vessel from a Burial at Auroville, Puducherry

Iron Technology and Metal Tools

The megalithic burials have abundant iron objects placed in the burials as grave goods. Weapons such as swords and daggers, axes, chisels, lamps and tripod stands were also found. Some of these objects were hafted to wooden or bone or horn handles and used. The iron tools were used for agriculture, hunting, gathering and in battles. Bronze bowls, vessels with stylish finials decorated with animals and birds, bronze mirrors and bells have also been found.

Recap

- The history of humans is intimately linked with the history of the earth. The earth originated around 4.54 billion years ago.
- The ancestors of human called hominins appeared about 5–7 million years ago.
- Although people gave divine explanations for the origin of humans, science believes in the theory of human evolution from the great apes.
- Humans began to domesticate animals and cultivate crops. The agricultural revolution led to many changes. Humans lived in permanent houses,

made pottery and with the surplus production, they developed various crafts.

- The earliest evidence of humans is available in Tamil Nadu around 2-1.5 million years ago.
- The Middle Paleolithic Culture is found in some parts of Tamil Nadu.
- The Mesolithic people lived in all the areas of Tamil Nadu.
- Neolithic culture is limited to northwestern part of Tamil Nadu.
- The Iron Age saw further expansion of people in various cultural zones. The foundation of subsequent Sangam Age was laid during this age.
- Iron tools were used in agriculture.



EXERCISE



I. Choose the correct answer

1. _____ is genetically closest to humans
(a) Gorilla (b) Chimpanzee
(c) Orang-utan (d) Great Apes



2. The period called _____ marks the beginning of agriculture and animal domestication.

- (a) Paleolithic (b) Mesolithic
(c) Neolithic (d) Megalithic

3. Direct ancestor of modern man was _____.

- (a) Homo habilis (b) Homo erectus
(c) Homo sapiens (d) Neanderthal man

4. _____ refers to the area covering Egypt, Israel-Palestine and Iraq.

- (a) Great Rift Valley
(b) Fertile Crescent
(c) Solo river
(d) Neander Valley

5. Sir Robert Bruce Foote, a geologist from England first discovered the _____ tools at Pallavaram near Chennai.

- (a) Microlithic (b) Paleolithic
(c) Mesolithic (d) Neolithic

6. (i) The period before the introduction of writing is called pre-history.

(ii) The pre-historic people developed language, made beautiful paintings and artefacts.

(iii) The pre-historic societies are treated as literate.

(iv) The pre-historic period is called ancient.

- a. (i) is correct
b. (i) and (ii) are correct
c. (i) and (iv) are correct
d. (ii) and (iii) are correct

7. (i) The Neolithic people used polished stone axes called Celts

(ii) Evidence of Neolithic village is found at Payyampalli in Chennai district

(iii) The cultural period that succeeded the Neolithic is called the Bronze Age

(iv) The period that witnessed domestication of animals and cultivation of crops is called Mesolithic

- a) (i) is correct
b) (ii) is correct
c) (ii) and (iii) are correct
d) (iv) is correct

8. **Assertion (A):** Many of the Mesolithic sites are founds nearby rivers and tanks.
Reason (R): Irrigation management developed during Mesolithic period.

- a) A and R are correct and R explains A
b) A and R are correct but R doesn't explain A
c) A is correct but R is incorrect
d) A and R both are incorrect

II. Fill in the blanks

- Hand axes and cleavers are the important tool types of the _____ culture.
- The methods and techniques involved in the production of stone tools are called _____ technology.
- _____ is known as the Middle Stone Age, as it is placed between the Paleolithic and Neolithic.

III. Find out the correct statement

- a) The concept 'survival of the fittest' contributed to the scientific understanding of human origins.
b) The book "On the Origin of Species" was published by Herbert Spencer.
c) Charles Darwin's theory of biological evolution connects with the process of natural selection.
d) Geology is the study of lithic technology.



2. a) Among the Great Apes Orang utan is genetically the closest to humans.
- b) The ancestors to humans were called Hominins and their origins have been traced to Africa.
- c) Flake cannot be used for tool making.
- d) Acheulian is the main block of stone from which small chips are flaked by using a hammer stone.

IV. Match the following

- | | |
|----------------------------------|------------------------------------|
| 1. Palaeo anthropology | - Teris |
| 2. Hand axe tools | - Venus |
| 3. Images on stone and bone | - Acheulian |
| 4. Red sand dunes | - Microliths |
| 5. Stone artefacts of small size | - The study of the human ancestors |

V. Answer the following briefly

1. Discuss how the age of speculation made humans become conscious and knowledgeable.
2. Write a note on the impact of pastoralism on the prehistoric people in Tamil Nadu.
3. List out the features of Megalithic Burial types.
4. Examine the tool making technical skills of lower Paleolithic people.

VI. Answer all the questions given under each caption

1. The developments in the fields of agriculture, pottery and metal tools are considered a landmark in the life of Megalithic period - Substantiate.
2. The history of humans is closely related to the history of the earth. Elucidate.

FUN WITH HISTORY

Student Activities

Mark the prehistoric sites on the world map

Organize an exhibition on the pre-historic sites of Tamil Nadu

Assignment with teacher's guidance

A power-point presentation on the origin of human life

A power-point presentation on the pre-historic tools

A power-point presentation on the scripts of the ancient period



REFERENCE BOOKS

1. Noboru Karashima, *A Concise History of South India Issues and Interpretations*. Oxford.
2. K.Rajan, *Iron Age-Early Historic Transition in South India: An Appraisal*. Padmashri Amalananda Ghosh Memorial Lecture, New Delhi: Institute of Archaeology.
3. Ralph, Burns and others. *World Civilizations (Vol. 1)*.



INTERNET RESOURCES

<http://www.sharmaheritage.com>

<https://www.nature.com>

<http://www.ancient-origins.net>

<http://humanorigins.si.edu>

<https://www.britannica.com>



ICT CORNER

Explore Pre-Historic Objects in Museums

Steps

- Scan the QR code and install the app.
- You can see three bars at the left side of the screen. Click them.
- When we click on 'collections', you can find world famous Museums. Select 'British Museum' Take a tour by clicking the yellow man icon. Click on 'Collections' to view the images of various objects in the Museum with high resolution and at the relevant ages.
- Click on the 'clock' to watch the timeline.

Back in Time



Website URL:

<https://play.google.com/store/apps/details?id=com.google.android.apps.cultural>

